Optimal Reconstruction and the Principal Subspace

Spectral Stability of Equity Covariance

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Abstract

Principal component analysis selects a low-dimensional subspace by maximizing retained variance, equivalently minimizing squared reconstruction error. We derive the reconstruction identity for an arbitrary orthogonal projector and express the retained variance through the sample covariance matrix. An eigenbasis argument then establishes optimality of the leading k-dimensional eigenspace. Finally, a strict boundary eigengap is shown to guarantee uniqueness of the optimal subspace.

1 Orthogonal projections and reconstruction error

Let $X \in \mathbb{R}^{n\times d}$ be a centered data matrix with rows x_i^T, where $x_i \in \mathbb{R}^d$ and $\sum_{i=1}^n x_i = 0$. We use the empirical covariance convention

$$\widehat{\Sigma} = \frac{1}{n} X^\mathsf{T} X \in \mathbb{R}^{d\times d}. \tag{1}$$

Fix $1 \le k \le d$ and an arbitrary k-dimensional subspace $\mathcal{U} \subseteq \mathbb{R}^d$ with orthonormal basis $u_1, \dots, u_k$. Write

$$U = [u_1, \dots, u_k] \in \mathbb{R}^{d\times k}, \qquad U^\mathsf{T} U = I_k, \qquad P_{\mathcal{U}} = UU^\mathsf{T}. \tag{2}$$

The matrix $P_{\mathcal{U}}$ is symmetric and idempotent, hence is the orthogonal projector onto $\mathcal{U}$.

Proposition 1.1 (Reconstruction and retained variance)**.** *For the projected observations $P_{\mathcal{U}} x_i$ and their reconstruction $XP_{\mathcal{U}}$,*

$$\boxed{\|X - XP_{\mathcal{U}}\|_F^2 = \|X\|_F^2 - \|XU\|_F^2 = \|X\|_F^2 - n\,\operatorname{tr}(U^\mathsf{T}\widehat{\Sigma}U).} \tag{3}$$

Proof. For each observation, decompose

$$x_i = P_{\mathcal{U}} x_i + (I_d - P_{\mathcal{U}})x_i.$$

The two components are orthogonal. The Pythagorean identity therefore gives

$$\|x_i - P_{\mathcal{U}} x_i\|^2 = \|x_i\|^2 - \|P_{\mathcal{U}} x_i\|^2. \tag{4}$$

Summing over all n observations and using the definition of the Frobenius norm,

$$\|X - XP_{\mathcal{U}}\|_F^2 = \sum_{i=1}^n \|x_i\|^2 - \sum_{i=1}^n \|P_{\mathcal{U}} x_i\|^2. \tag{5}$$

Since $P_{\mathcal{U}}^\mathsf{T} P_{\mathcal{U}} = P_{\mathcal{U}}$,

$$\begin{aligned}\sum_{i=1}^n \|P_{\mathcal{U}} x_i\|^2 &= \sum_{i=1}^n x_i^\mathsf{T} UU^\mathsf{T} x_i = \sum_{i=1}^n \|U^\mathsf{T} x_i\|^2 \\ &= \|XU\|_F^2.\end{aligned}$$

Moreover, writing $U = [u_1, \dots, u_k]$,

$$\begin{aligned}\|XU\|_F^2 &= \sum_{j=1}^{k} \|Xu_j\|^2 = \sum_{j=1}^{k} u_j^\mathsf{T} X^\mathsf{T} X u_j \\ &= n \sum_{j=1}^{k} u_j^\mathsf{T} \widehat{\Sigma} u_j = n \operatorname{tr}(U^\mathsf{T} \widehat{\Sigma} U). \end{aligned} \tag{6}$$

Substituting into (5) proves (3). □

Because $\|X\|_F^2$ is independent of $\mathcal{U}$, minimizing squared reconstruction error over k-dimensional subspaces is equivalent to maximizing

$$\operatorname{tr}(U^\mathsf{T} \widehat{\Sigma} U) \quad \text{subject to} \quad U^\mathsf{T} U = I_k. \tag{7}$$

2 Why the leading eigenspace is optimal

Let $v_1, \dots, v_d$ be an orthonormal eigenbasis of the symmetric positive semidefinite matrix $\widehat{\Sigma}$, with corresponding eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0, \qquad \widehat{\Sigma} v_i = \lambda_i v_i.$$

Theorem 2.1 (Maximum retained variance)**.** *For every $U \in \mathbb{R}^{d \times k}$ with $U^\mathsf{T} U = I_k$,*

$$\boxed{\operatorname{tr}(U^\mathsf{T} \widehat{\Sigma} U) \leq \sum_{i=1}^{k} \lambda_i.} \tag{8}$$

Equality is attained by $U = [v_1, \dots, v_k]$. Thus the leading k-dimensional eigenspace minimizes the squared reconstruction error.

Proof. Express each column u_j in the eigenbasis,

$$u_j = \sum_{i=1}^{d} a_{ij} v_i, \qquad a_{ij} = v_i^\mathsf{T} u_j. \tag{9}$$

Because the eigenvectors are orthonormal,

$$u_j^\mathsf{T} \widehat{\Sigma} u_j = \sum_{i=1}^{d} \lambda_i a_{ij}^2.$$

Summing over j yields

$$\operatorname{tr}(U^\mathsf{T} \widehat{\Sigma} U) = \sum_{i=1}^{d} \lambda_i w_i, \qquad w_i := \sum_{j=1}^{k} a_{ij}^2 = \|P_{\mathcal{U}} v_i\|^2. \tag{10}$$

As an orthogonal projection cannot increase Euclidean norm, $0 \leq w_i \leq 1$. Furthermore, Parseval's identity gives

$$\sum_{i=1}^{d} w_i = \sum_{j=1}^{k} \sum_{i=1}^{d} (v_i^\mathsf{T} u_j)^2 = \sum_{j=1}^{k} \|u_j\|^2 = k. \tag{11}$$

Using $\lambda_i \leq \lambda_k$ when $i > k$ and (11),

$$\begin{aligned}\sum_{i=1}^{d} \lambda_i w_i &\leq \sum_{i=1}^{k} \lambda_i w_i + \lambda_k \sum_{i=k+1}^{d} w_i \\ &= \sum_{i=1}^{k} \big[\lambda_i w_i + \lambda_k (1 - w_i)\big] \\ &\leq \sum_{i=1}^{k} \lambda_i, \end{aligned} \tag{12}$$

where the final inequality uses $\lambda_i \geq \lambda_k$ and $0 \leq w_i \leq 1$ for $i \leq k$. Finally, take $U = [v_1, \ldots, v_k]$. Then $w_i = 1$ for $i \leq k$ and $w_i = 0$ otherwise, so equality holds. □

$$\boxed{\min_{U^\mathsf{T}U = I_k} \|X - XUU^\mathsf{T}\|_F^2 = \|X\|_F^2 - n\sum_{i=1}^{k} \lambda_i = n\sum_{i=k+1}^{d} \lambda_i.} \tag{13}$$

Here $\|X\|_F^2 = n \operatorname{tr}(\widehat{\Sigma}) = n\sum_{i=1}^{d} \lambda_i$; the final equality is simply a rearrangement of the retained-variance identity.

3 Uniqueness and the boundary eigengap

A maximizer need not be unique when the covariance eigenvalues are tied at the boundary of the selected subspace. The following argument specifies when the optimal subspace is unique.

Theorem 3.1 (Uniqueness of the optimal subspace)**.** *Suppose $1 \leq k < d$ and $\lambda_k > \lambda_{k+1}$. Then the unique k-dimensional subspace maximizing* (7) *is*

$$\mathcal{U}_\star = \operatorname{span}\{v_1, \ldots, v_k\}.$$

The choice of orthonormal basis within $\mathcal{U}_\star$ remains nonunique.

Proof. With the weights w_i from (10), set

$$T := \sum_{i=k+1}^{d} w_i = \sum_{i=1}^{k} (1 - w_i) \geq 0. \tag{14}$$

The difference between the maximal and the retained variance satisfies

$$\begin{aligned} \sum_{i=1}^{k} \lambda_i - \operatorname{tr}(U^\mathsf{T}\widehat{\Sigma}U) &= \sum_{i=1}^{k} \lambda_i(1 - w_i) - \sum_{i=k+1}^{d} \lambda_i w_i \\ &\geq \lambda_k \sum_{i=1}^{k} (1 - w_i) - \lambda_{k+1} \sum_{i=k+1}^{d} w_i \\ &= (\lambda_k - \lambda_{k+1})T. \end{aligned} \tag{15}$$

If U is optimal, the left side is zero. The strict eigengap forces $T = 0$. As each $w_i \geq 0$, this implies $w_i = 0$ for all $i > k$. By (10),

$$\|P_{\mathcal{U}} v_i\|^2 = 0 \quad (i > k),$$

so every vector in $\mathcal{U}$ is orthogonal to $v_{k+1}, \ldots, v_d$. Consequently,

$$\mathcal{U} \subseteq \operatorname{span}\{v_1, \ldots, v_k\}.$$

Both subspaces have dimension k; hence they coincide. □

The boundary eigengap $\lambda_k - \lambda_{k+1}$ thus controls uniqueness of the optimizing *subspace*, not uniqueness of its orthonormal basis. This distinction is essential when comparing estimated PCA subspaces across samples or time windows.